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A kirigami-metamaterial triboelectric nanogenerator for real-time wave load monitoring and early warning in marine structures

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ABSTRACT

Real-time monitoring and early warning of long-term wave loads on marine structures are critical for preventing catastrophic damage. Traditional sensors, however, struggle with large-deformation tolerance, dynamic response, and self-powered operation, especially in shallow, typhoon-prone waters where wave actions are highly stochastic. This study presents a kirigami-metamaterial based triboelectric nanogenerator (KM-TENG) that overcomes these limitations through innovative mechanical design. The compact cylindrical three-layer structure incorporates a programmable sensitive kirigami element, providing 125% deformation capacity and tunable stiffness via geometric parameters. By combining a waterproof layer for load transfer and a rigid conversion unit for triboelectric coupling, the KM-TENG produces electrical signals via a sliding-mode mechanism enabled by nested multilayer enclosure walls within the kirigami component. This design allows simultaneous detection of both point-contact mechanical impacts and face-contact fluid pressures, and has been validated under broad temperatures, salinity ranges and other demanding conditions, including temperature (11.3-32.1 °C), salinity (up to 35‰), and sediment loads (15 kg/m^3). The system offers an integrated real-time solution incorporating data acquisition, LoRa wireless transmission, and threshold-based warning algorithms for wave pressure monitoring. This work provides a structural innovation for self-powered sensing in challenging marine settings and establishes a practical framework for risk mitigation of marine structures.

1. Introduction

In recent years, marine structures have been trending towards larger and more complex designs, encompassing seacrossing bridges, offshore wind turbines, and oil drilling platforms [1–4]. Unlike terrestrial structures, marine structures are prone to damage from long-term random wave loading and face the risk of catastrophic events initiated by abnormal wave heights and strong nonlinear wave impacts. Evidently, real-time monitoring of structural loading is of paramount importance to ensure safety. Dynamic wave pressure, a key indicator of structural load, is essential for safety assessments. However, the high randomness of waves and the uncertainty of dynamic wave pressures present substantial challenges, particularly in shallow nearshore waters and during typhoons [5–7]. Extreme waves can generate intense instantaneous impact loads with pronounced pulsating characteristics. Conventional pressure sensors, such as resistive [8], capacitive [9,10], and fiber-optic sensors [11], exhibit limitations in terms of adaptability, deformation resistance, and rapid response, making them less effective in such conditions. Moreover, their dependence on external power and wired data transmission can severely compromise real-time monitoring capabilities in harsh environments. Therefore, there is a pressing need for self-powered, easily deployable, adaptable, deformation resistant, and fast-responding dynamic pressure sensors, as well as real-time wireless monitoring systems, to enhance the safety of marine structures [12,13].

Triboelectric nanogenerators (TENGs) are considered a promising solution due to their low cost, wide variety of material choices, and flexible structural design [14–17]. They can convert mechanical energy into electrical energy, enabling self-powered sensors to correlate electrical signals with changing physical quantities. Notably, the self-powered advantage of TENGs has been fully validated in diverse monitoring scenarios, laying a solid foundation for their practical application in unattended and remote environments [18–20]. Many studies collectively confirm that TENGs, with their prominent self-powered feature and strong environmental adaptability, are well-suited for developing efficient and sustainable monitoring systems [21–23]. In marine environmental monitoring, TENGs have been developed into various wave sensors, such as those for wave height, period, frequency, speed,

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length, steepness, and wave force, and have been validated in practical applications [24–32]. Many reported hydrostatic pressure sensors can detect water levels in undulating waves sensitively and are composed of elements such as floating rods and floating balls [25,27,29]. These elements move with free surface fluctuations, forming contact-sliding mode TENGs. However, because of the presence of marine structures, the dynamic wave pressures induced by drag and inertial forces differ significantly from those in independent pressure fields. The placement and fixation of these sensors is not suitable for monitoring wave pressures on marine structure surfaces. Although a series of high-performance sensors have been developed to monitor forces and pressures, some lack waterproofing and are not easily applicable underwater [33–36], others lack adjustability of measurement range and sensitivity [37,38], thus limiting their flexibility in various application scenarios. Most importantly, current TENG architectures face two fundamental limitations: (1) an inability to directly measure the wave loads experienced by marine structures due to structural incompatibility, and (2) insufficient flexibility in adjusting measurement range and sensitivity to accommodate diverse marine conditions. These structural deficiencies significantly impede accurate assessment of wave loads on marine infrastructure. Thus, designing TENGs that meet monitoring requirements for marine structures remains challenging but necessary. Fortunately, mechanical metamaterials offer valuable insight. Many TENGs have incorporated designs from mechanical metamaterials, such as kirigami metamaterials [39–41], origami metamaterials [42–44], chiral metamaterials [45] and compression-torsion metamaterials [46]. These designs leverage the advantages of TENGs and the enhanced deformation resistance, excellent recoverability, and negative Poisson's ratio of mechanical metamaterials, achieving remarkable results. However, their application has largely been confined to controlled environments. The integration of metamaterial-based TENGs into functional, waterproof systems for the direct, quantitative measurement of wave loads on actual marine structures has not been achieved, particularly when facing the combined challenges of large deformations and suspended sediment. This represents a critical barrier to their practical deployment.

This study proposes a self-powered wireless monitoring system based on a kirigami-metamaterial triboelectric nanogenerator (KM-TENG) to overcome critical limitations of conventional marine structural sensors. These limitations include inadequate tolerance for large deformations, insufficient dynamic response, reliance on external power supplies, and dependence on cumbersome cabling, particularly under the highly random and extreme wave loads prevalent in shallow and typhoon-prone waters. The core of the system, the KM-TENG, features a programmable kirigami structure as its sensitive element, enabling exceptional deformation capacity and tunable stiffness to reliably capture stochastic wave impacts. The functional waterproof packaging ensures robust operation in harsh marine environments. Crucially, this integrated system combines real-time sensing, self-powered operation, LoRa wireless data transmission, and intelligent threshold-based warnings, providing a practical solution for proactive structural risk management.

2. Materials and methods

2.1. Conceptualization, structure and working mechanism

KM-TENG can play a crucial role in monitoring wave loads for marine structures (Fig. 1a). Under normal and extreme weather conditions, long-term wave loading poses an escalating threat to marine structures, including offshore wind turbines and sea-crossing bridges. These loads directly induce fractures and damages to structural foundations, thus threatening the safety and stable operation of engineering systems. Given that dynamic wave pressure, especially wave slamming pressure, serves as a critical monitoring indicator, the KM-TENG is developed in this study to address this need. Designed to be deployed as data acquisition modules, these devices can be arranged in arrays, linearly along the seabed and circumferentially around the surface of marine structures. They form a network integrated with data transmission units, enabling wireless data transfer to the server terminal. This configuration facilitates the real-time monitoring of wave loading on marine structures.

Fig. 1b illustrates the three-layer integrated compact structure of the KM-TENG, which sequentially consists of the waterproof layer, the sensitive element, and the load-electricity conversion component. The key mechanical properties of all constituent materials are summarized in Table S1. Physical dimensions and images of a single KM-TENG are also presented in Note S1. The waterproof layer, made of silicone rubber A (Young's modulus < 1 MPa), serves as the outermost barrier against water intrusion while efficiently transferring external loads. The sensitive element, composed of silicone rubber B with higher Young's modulus, features a kirigami design with nested enclosure walls. It forms a guided sliding system to match the corresponding part of the load-electricity conversion component, enabling programmable mechanical responses under load that are distinct from conventional elastomers. Unlike standard materials, the precise geometric pairing of the kirigami structure allows for controlled large deformations, critical for capturing transient wave impacts. The load-electricity conversion component, fabricated from rigid PLA, provides structural reinforcement and cooperates with the sensitive element to facilitate charge generation through the guided sliding motion. The KM-TENG is fabricated for marine wave load monitoring, as detailed in Section 2.2.

Fig. 1c shows the internal structure of the KM-TENG. The concentric annular corrugations belong to the waterproof layer, while the sensitive element is composed of a kirigami part (circular diaphragm with hexagonal notches) and its own enclosure wall part (denoted as multilayer hexagonal enclosure walls Part I). Besides, the load-electricity conversion component also contains a corresponding enclosure wall part (multilayer hexagonal enclosure walls Part II). When an external load is transmitted to the concentric annular corrugations of the waterproof layer, its annular corrugations deform in synergy, causing the membrane to concave inward. That drives the kirigami part to buckle at the hexagonal notches. This buckling motion induces the enclosure wall part I to slide within relative enclosure wall part II. The enclosure walls of the conversion component match the height and number of layers of the sensitive element but have slightly larger inter-wall spacing, allowing relative sliding under loads. This sliding mechanism causes the Nylon membrane (attached to the enclosure wall part I) and PTFE membrane (attached to the enclosure wall part II) to periodically contact and separate, generating alternating charges due to electronegativity differences—PTFE gains electrons (negative charge), and Nylon loses electrons (positive charge). As the load recedes, the kirigami part in the sensitive element restores the structure, prompting electron transfer in the external circuit and forming a periodic voltage signal. The sequential stages of this sliding mechanism and charge transfer process are visually detailed in Fig. 1d.

2.2. Fabrication of KM-TENG

The load-electricity conversion component is 3D printed using black PLA. The model is designed in SolidWorks2021 and printed layer by layer. The waterproof layer and sensitive element are produced via vacuum casting with three liquids (Liquid A, B, C). For the waterproof layer, the mixing ratio is 100:100:400. For the sensitive element, it is 100:100:100 (Shore 70 A) and 100:100:300 (Shore 40 A). After mixing and degassing, the mixture is poured into molds and cured at 70 °C for 60 min.

The multilayer hexagonal walls of the sensitive element are unfolded into a plane sketch in SolidWorks2021. A 50 μm nylon film (PA 66) and a 100 μm double-sided adhesive aluminum foil are cut according to the plane sketch and laminated together. Then they are bonded to the surface of enclosure wall part I. In addition, an 80

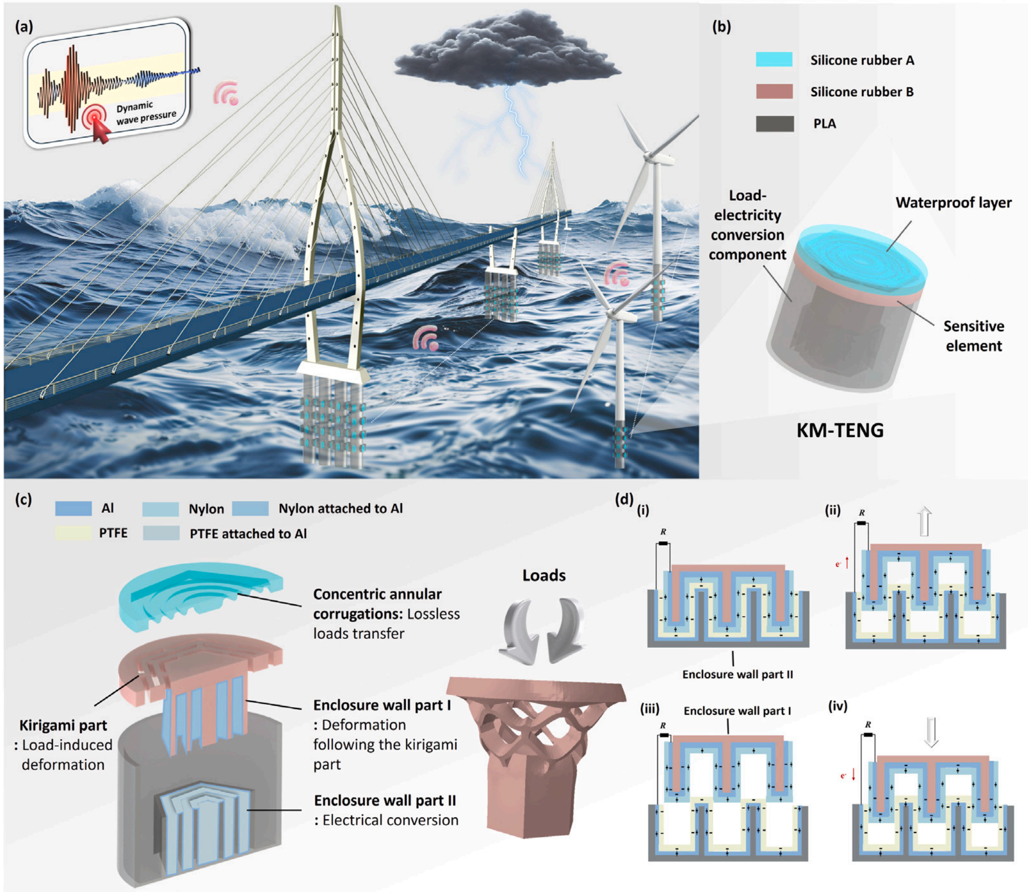

Fig. 1. Application scenarios and structural design of the KM-TENG. (a) Potential application scenarios. (b) Structure schematic of the KM-TENG. (c) Internal materials and key architectures of the KM-TENG. (d) The working principle of the KM-TENG.

μm single-sided adhesive aluminum foil is cut with a 50 μm PTFE film and attached to the enclosure wall part II. Nylon (PA 66) is selected as the positive material due to its strong electron-donating tendency, which, when paired with PTFE, yields the highest and most stable open-circuit voltage compared to several commonly used positive triboelectric materials (including PET, PP, and PU). PTFE is selected as the negative material as a result of its strong electron affinity and its superior performance. Their availability as a flexible film also ensures both high triboelectric performance and straightforward integration with the structure of KM-TENG. The waterproof layer and the sensitive element are bonded together. Finally, 1 mm silicone adhesive is applied to the KM-TENG exterior and cured for 24 h.

2.3. Mechanical experiment and numerical simulation

The mechanical characterization of the sensitive element of the KM-TENG is carried out by utilizing a microcomputer-controlled electronic loading system (produced by Shandong Hanshen Fatigue Testing Machine Co., Ltd.). First, a 3D-printed PLA substrate is fixed to the base of the loading machine, which enables the observation of the deformation of the sensitive element. Then, the sensitive element is attached to the substrate and brought into contact with the loading tool, which is fixed to the gripper clamp of the machine. The loading tool deforms the sensitive element at a constant rate through the controlled movement of the gripper clamp. Moreover, the mechanical characterization, mechanical programmability of the sensitive element, and the influence of the waterproof layer are analyzed by Abaqus2022. Further detailed information is provided in Note S3, Note S4 and Note S7.

2.4. Electrical experiment

To investigate the output performance of KM-TENG in both air and water, an integrated servo motor system (60A6S-M01330-B) with force feedback (JLBU-1-50KG) is designed. This system regulates the loading displacement, frequency, and cycle count. A small acrylic water tank (194 mm × 194 mm × 100 mm) is placed beneath the system. The KM-TENG is fixed at the bottom of the tank, ensuring it can be effectively exposed to vertical point-contact mechanical impact during the experiment. The servo motor deforms the KM-TENG via a loading tool. The small cross-section of the cylindrical loading tool ensures the contact condition approximates an ideal point contact. Meanwhile, a force sensor, installed between the servo motor and the loading tool, simultaneously records the magnitude of the applied point-contact mechanical impact force and transmits real-time force data to a computer via a data acquisition system (DAQ). The KM-TENG is connected to an oscilloscope to generate waveform signals and measure open-circuit voltage (V_{oc}). By varying the loading parameters (displacement, frequency), a range of impact forces and corresponding voltage outputs is generated. The peak force values from the force sensor and the peak voltage values from the oscilloscope are then correlated to establish a linear calibration curve. This setup enables electrical performance evaluation in both air and water environments.

A dynamic air pressure generation device is designed to evaluate the response of KM-TENG to fluid pressure. The device consists of a stable pressure source, chamber, exhaust port, air inlet, commercial pressure sensor, and circulating air pump. The pressure source, which is an oil-free air compressor, supplies regulated gas flow to the circulating air pump through a valve. The circulating air pump serves as a transfer hub, periodically filling the chamber with gas, thus creating cyclical pressure changes. By adjusting delivery and pause times of the pump, the air pressure frequency can be modified, while the valve opening regulates the maximum pressure. The KM-TENG, which is fixed at the bottom of the chamber, directly detects these pressure variations. This configuration ensures the application of face-contact pressures across the KM-TENG's surface. A commercial pressure sensor continuously monitors chamber pressure, which is referenced to atmospheric pressure through the exhaust port. This sensor provides reference pressure values that are synchronized with the electrical output of the KM-TENG. The periodic deformation of KM-TENG generates corresponding electrical signals. The KM-TENG is also connected to an oscilloscope to capture waveform signals and measure V_{oc}. By correlating the peak pressure values from the reference sensor with the peak voltage values from the KM-TENG, a linear relationship is established for fluid pressure calibration.

Output voltage signals are obtained through an oscilloscope (DS1202Z, RIGOL, China) that is equipped with a high-impedance probe (UT-P12, 100 MHz), and V_{oc} is measured in this process.

2.5. Wireless communication module

The raw input signals from the KM-TENG are acquired and conditioned using a 24-bit sigma-delta ADC (CS1238), which integrates programmable gain amplification (PGA: 1–128×) and anti-aliasing filtering to prevent high-frequency noise from corrupting digitization and to preserve signal integrity. Besides, the CS1238 minimizes the need for external active components, reducing additional current draw. The digitized data streams are processed by an STM32F103C8T6 ARM Cortex-M3 microcontroller, which applies integrated digital filtering to attenuate high-frequency noise while retaining essential signal components, serving as a foundational noise reduction step prior to transmission. For reliable long-range wireless communication, a WH-L101-L LoRa module based on the Semtech SX1268 chip is employed. This module utilizes spread spectrum technology to provide inherent processing gain against narrowband interference, ensuring robust data transfer in dynamic environments. The module operates in the 410–525 MHz frequency band (default 470 MHz), supports an operating voltage range of 1.9–3.7 V, and features a low hibernation current of 2.5 μA. Continuous system operation is ensured by a 5 V lithium polymer battery pack with a capacity of 3400 mAh. Shielded cabling is used between KM-TENG and the acquisition module to minimize electromagnetic interference and maintain signal fidelity during transmission. Additionally, a MATLAB-based host computer implements data-adaptive Wiener filtering to further enhance signal quality during offline analysis.

3. Results and discussion

3.1. Mechanical characterization

The sensitive element of the KM-TENG is the load-sensitive module, with its kirigami-structured diaphragm acting as the primary load-bearing component. Investigating the mechanical behavior of the kirigami part is crucial for understanding the overall mechanical properties of the KM-TENG. Initially, three kinds of kirigami parts are conceived and compared: vertically extensible, radially symmetric triangle-kirigami, quadrilateral-kirigami, and hexagonal-kirigami structures (Fig. 2a). These polygon-kirigami configurations are created by introducing precisely patterned notches into two-dimensional circular diaphragm, resulting in periodic hollowed-out regions. The geometric construction begins with a central polygon, from which concentric rings of polygons extend outward. The midpoints of the innermost polygon's sides are connected to the first outer ring via side bonds of uniform width, while the remaining areas form notches. Similarly, the vertices of the first outer ring are connected to the second outer ring via vertex bonds of consistent width, with the remaining areas again forming notches. This hierarchical pattern repeats outward, creating a self-similar kirigami structure.

To investigate the impact of polygon shape on mechanical characterization, the basic parameters of the three polygon-kirigami configurations are kept consistent. The outer radius R of the central complete polygon, the distance d between adjacent vertices of concentric ring-shaped polygons, and the width b of the side and vertex bonds are maintained at the same values. Additionally, the number of closed trapezoidal areas n, the thickness t of the kirigami part, the length L of the polygon sides, the length ω_1 of the side bonds, the distance ω_2 between adjacent solid polygon sides, and the Shore A hardness remain unchanged. Due to the fixed parameter d, the dimensions of L, ω_1 and ω_2 differ across the different polygon shapes. Initial design estimation of geometric parameters is based on theoretical model, size requirements and empirical principles. A detailed list of the structural parameters used in this study is provided in Note S2. The simplified deformation of the polygon-kirigami structure, as illustrated in the side view in Fig. 2a, clearly shows the structural response to external loading. When subjected to a vertical downward force, the kirigami structure deforms, with the total deformation denoted as D. During this process, each closed trapezoidal area within the kirigami structure deforms, and the areas closer to the central polygon exhibit more significant deformation. This behavior can be attributed to the inherent design of the structure, which can be modeled as a beam system. Specifically, the structure can be approximated as a beam model composed of beam elements located between the vertex bonds and side bonds. This configuration allows the kirigami structure to be equivalently represented as two cantilever beam models, each with one end experiencing a bending force. This beam model analogy offers a useful framework for analyzing the mechanical behavior of the polygon-kirigami structure under loads [47–49]. To fully characterize the actual mechanical performance of the polygon-kirigami structures under practical loading conditions and validate the rationality of the initial design, the macroscopic experiments and advanced large-deformation finite element method (FEM) simulations are conducted.

The mechanical performance of prototypes fabricated with these parameters is then experimentally characterized, providing essential baseline data on the structure's deformation behavior and stiffness characteristics. The loading machine is used to characterize the loading and unloading of the sensitive element combined with three polygon-kirigami structure under static load. At a constant loading rate of 10 mm/min, Fig. 2b-d present the loading and unloading loops for each kirigami geometry at five different maximum deformations, i.e., D_{max} = 5 mm, 10 mm, 15 mm, 20 mm, 25 mm, respectively. The upper curve represents loading, while the lower curve represents unloading. Although these curves do not overlap, they form a closed hysteresis loop, which reflects the elastic lag of material and structure and dissipated energy during loading and unloading. Fig. 2e compares force–displacement relationships of the three polygon-kirigami structures at a maximum deformation of 25 mm. During the initial stage, the force–displacement curves for all three polygon-kirigami patterns show a linear increase at a relatively low rate. Once the displacement exceeds 14.1 mm, the hexagon-kirigami curve increases rapidly while maintaining an approximately linear trend. The growth rate jumps from 0.78 N/mm to 1.63 N/mm. In contrast, the other two polygon-kirigami structures exhibit downward trend and negative stiffness behavior beyond their respective first peaks (D = 15.4 mm for quadrilateral-kirigami and D = 18.1 mm for triangular-kirigami). This is attributed to the coupling deformation between internal and external beam elements under compressive loading. After the transformation of internal and external deformations, the curves rise again from the second turning point (D = 19.3 mm for quadrilateral-kirigami, D = 20.6 mm for triangular-kirigami), with growth rates exceeding those of the initial stage. Although negative stiffness is observed as an interesting phenomenon for energy absorption or meta-material

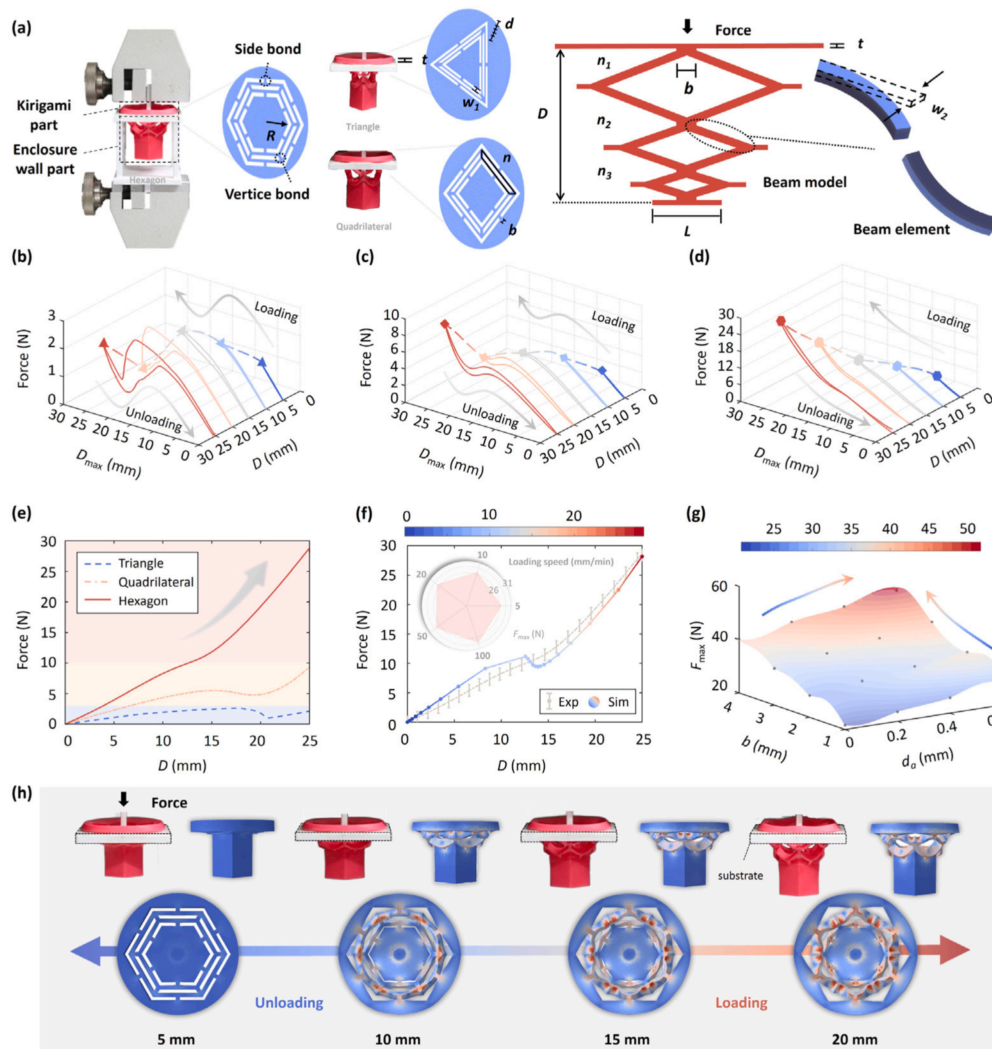

Fig. 2. Mechanical characterization of the sensitive element. (a) Sensitive elements with three kinds of kirigami parts considered in this study and simplified structure deformation. (b) Loading and unloading process of triangle-kirigami with different maximum deformation. (c) Loading and unloading process of quadrilateral-kirigami with different maximum deformation. (d) Loading and unloading process of hexagon-kirigami with different maximum deformation. (e) Force–displacement curves of the upper part with three kirigami shapes when the maximum deformation is 25 mm. (f) Force–displacement curve comparison of the hexagon-kirigami between the experiment and numerical simulation. (g) F_{max} of the hexagon-kirigami with different structural parameters b and d_a in the numerical simulation. (h) Comparison of the hexagon-kirigami deformation in the experiment and numerical simulation.

applications, it is undesirable in the sensing context due to its non-monotonic force–displacement relationship, which would complicate signal interpretation and sensor calibration. In contrast, the hexagon-based design provides the widest stiffness tunability range along with a high stretching ratio of 125% and excellent recovery (99.8%), making it ideally suited for robust and predictable sensor performance.

To further investigate the mechanical characterization of the hexagon-kirigami structure, a numerical model is developed (detailed calculation settings are provided in Note S3). Comparisons between experimental and numerical simulation results for deformation and force–displacement relationships in hexagon-kirigami are shown in Fig. 2f and 2h. The force–displacement curve trends from the simulations in Fig. 2f align with the experimental data, and some simulated results closely match the experimental values, validating the reliability of the numerical simulation. Additionally, measurements of F_{max} at different loading rates show that while faster rates produce larger F_{max} values, the differences are minimal between 5 mm/min and 100 mm/min. Deformation image comparisons between experiments and simulations, along with pressure cloud diagrams, are illustrated in Fig. 2h. The numerical simulation deformation process is consistent with key experimental observations. The pressure cloud diagram reveals that the highest structural stresses (indicated by colors approaching red) occur around the side and vertex bonds. Consequently, to quantitatively evaluate the influence of key geometric parameters on structural stiffness, a sensitivity analysis is conducted using the finite element model. The control variable method is used to isolate the effect of individual parameters, including the distance between adjacent vertices d, the width of the side and vertex bonds b, the structural thickness t, and the intrinsic hardness of the material. For each parameter, force–displacement curves are obtained under prescribed displacement loading, and F_{max} and the slope of the curve are extracted to assess the structural stiffness and load-bearing capacity. Notably, the analysis reveals distinct regulatory effects of different parameters: variations in t and the shore hardness of silicone rubber yield significant changes in both F_{max} and stiffness, while modifying b and d results in finer, incremental adjustments to these properties. The results confirm that each of these parameters significantly influences the mechanical response, enabling flexible modulation of the stiffness of the sensitive element. This mechanical programmability, which leverages the differential tuning effects of parameters, allows the measurement range and sensitivity of the KM-TENG to be tailored to specific application requirements. Detailed parameter settings and results are provided in Note S4.

To further evaluate the long-term durability of the sensitive element, a fatigue life prediction is conducted based on finite element analysis in Abaqus. The maximum principal strain value of 1.02, obtained at the critical location under 25 mm displacement amplitude (Fig. S7), is input to the established E-N curve for silicone rubber (Fig. S8) [50].This analysis predicts a fatigue life exceeding 10^5 cycles under maximum displacement conditions, which suggests a substantially longer service life during actual monitoring operations where deformation amplitudes are typically less than 25 mm. Detailed methodology and results are provided in Note S5.

3.2. Electrical performance

Dynamic wave pressure is challenging to detect and verify. To explore the electrical performance of KM-TENG under cyclic loading of water and air pressure, the servo motor force feedback system and dynamic air pressure generation device are employed, as shown in Fig. 3a and 3b. Based on the characteristics of vertical mechanical impact force (instantaneous and localized) and dynamic fluid pressure (continuous and distributed) and the discussion in Section 3.1, two sets of sensitive elements with different structural parameters are fabricated. For the condition of vertical mechanical impact force, the parameters are set to $b = 2$ mm, $d_a = 0$ mm, $t = 5$ mm, and the Shore A hardness is 70. For the dynamic fluid pressure condition, the values of b, d_a, and t remain unchanged, while the Shore A hardness is 40.

The generation of electrical signals depends on the coordinated interaction between the enclosure walls of the sensitive element and the load-electricity conversion component. To optimize the structural parameters of the multilayer hexagonal enclosure walls on the conversion component, systematic experiments are conducted using a servo motor force feedback system. Specifically, combinations of sensitive elements and conversion components with varying parameters are tested: layer numbers of 1, 2, and 3, and spacing distances of 0.8, 1.0, and 1.2 mm. Detailed optimized parameters are provided in Note S6. The effects of these parameters on open-circuit voltage V_{oc} at 2 Hz are shown in Fig. 3c and 3d. After low-pass filtering and adjustment of V_{oc} to a zero minimum, peak V_{oc} is observed to increase with downward deformation, consistent with the characteristics of contact-sliding mode TENGs. Among all combinations, the highest peak V_{oc} is yielded by the 3-layer configuration with 0.8 mm spacing. The observed performance trends can be attributed to the structural and operational mechanisms of the nested multilayer design. Performance enhancement with increasing layer number arises from two key effects: first, the effective triboelectric contact area within a finite volume is increased; second, each adjacent layer pair functions as an independent sliding-mode TENG unit, forming a series connection where additive voltage output is enabled. This series configuration contributes to a triboelectric coupling efficiency close to quadruple when the layer number is increased from 1 to 3. Regarding spacing distance, reducing it brings the walls of the sensitive element and conversion component closer, which theoretically improves voltage output by enhancing charge transfer but also increases friction—posing a risk of impeding smooth sliding. Notably, stiction between layers is caused by further reducing the spacing to 0.6 mm, which degrades performance. Thus, the structural parameters (number of layers and spacing distance) are optimized to balance electrical performance and operational reliability: 3 layers and 0.8 mm spacing are selected as the optimal combination, maximizing V_{oc} while ensuring stable sliding motion.

To adapt to high-humidity environments, additional waterproof measures are implemented. Firstly, a waterproof layer is designed to cover the sensitive element. Nonlinear multi-body simulations are introduced because of the contact behavior (friction, normal interaction) between the sensitive element and the waterproof layer. The structural parameters of waterproof layer and the numerical simulation settings used to evaluate its effectiveness are detailed in Note S7. Fig. 3e compares the force–displacement curves with and without the waterproof layer. The trends are consistent, with only the magnitude exhibiting variations. After deformation exceeds 7 mm, the force value with the waterproof layer is slightly higher. These results suggest that the waterproof layer retains good elasticity and effectively transmits the load. Secondly, the sides and base of the overall structure are encapsulated with silicone glue to ensure the complete sealing of the KM-TENG. The V_{oc} of the KM-TENG under vertical mechanical impact force is measured in air using the servo motor force feedback system. Fig. 3f displays the peak voltages at different deformations and frequencies, indicating that the peak voltages increase as the deformation increases. This effect is more significant at higher frequencies, which is attributed to the enhanced charge transfer rate and reduced charge dissipation at higher frequencies. The KM-TENG reaches a maximum peak V_{oc} of 40 V at deformation of 15 mm and frequency of 2 Hz. The error bars (Fig. S13) of output voltages of samples (sample size: 3) demonstrate the excellent reproducibility of KM-TENG. The electrical performance of the KM-TENG in water is also investigated. Fig. 3g, 3h, and 3i present the output voltages at different deformations in air, at different frequencies in water, and comparisons between them. The minimal shielding effect of water demonstrates the effectiveness of encapsulation and the ability of KM-TENG to operate in aqueous environments. In addition, the electrical output performance of the KM-TENG is systematically characterized under a constant loading frequency of 2 Hz and displacement of 9 mm. KM-TENG generates a short-circuit current (I_{sc}) up to 1 μA (Fig. S14b) and Q_{SC} (Fig. S14c) remains constant at approximately 20.56 nC. The relationship between output power and external load resistance is investigated (Fig. S14d), revealing that output power reaches its maximum at an optimal load resistance of 70 MΩ, demonstrating efficient impedance matching. At this point, the volumetric power density peaks at 34.82 mW/m^3 (Fig. S14f).

Given the differences between mechanical impact forces and fluid pressures, the dynamic air pressure generation device is used to verify the performance of KM-TENG under dynamic fluid pressure. Its core purpose is to simulate fluid pressures in general. The device design ensures correspondence with fluid pressure properties. Specifically, for key parameters in fluid pressure variations (e.g., wave frequency, which is a typical parameter in wave-related fluid pressures), modulation is achieved by adjusting the delivery and pause times of the integrated pump; for parameters reflecting pressure magnitude (e.g., wave amplitude), simulation is realized by regulating valve opening to control gas flow rate and thus maximum pressure. Furthermore, the chamber is connected to the atmosphere to prevent enclosed-space pressure distortion, maintaining consistency with real fluid pressure variations. Results in Fig. 3j reveal that the KM-TENG maintains a stable voltage output within the frequency range of 0.55 Hz to 1.67 Hz at a fixed pressure of 19.6 kPa. At a fixed frequency of 1 Hz, Fig. 3k illustrates a gradual increase in V_{oc} as the peak dynamic pressure rises. These findings confirm that the KM-TENG can operate effectively under vertical mechanical impact forces in both air and water, as well as under dynamic air pressures and dynamic wave pressures.

3.3. Mechanical-electrical response and robustness

To establish a robust correlation relationship between mechanical stimuli and electrical outputs, the analysis is anchored in experimentally validated data. Fig. 4a–b demonstrate that the peak output voltage maintains a linear correlation with vertical mechanical impact forces across deformation frequencies of 0.5 Hz, 1 Hz, 1.5 Hz, and 2 Hz, with an average coefficient of determination R^2=0.98. The sensitivity increases with frequency, reaching 0.82 V/N in air and 0.77 V/N in water at 2 Hz, a 45% enhancement from the 0.5 Hz sensitivity (0.56 V/N in air). This frequency dependence arises from the dynamic buckling behavior of the kirigami structure, higher frequencies enables

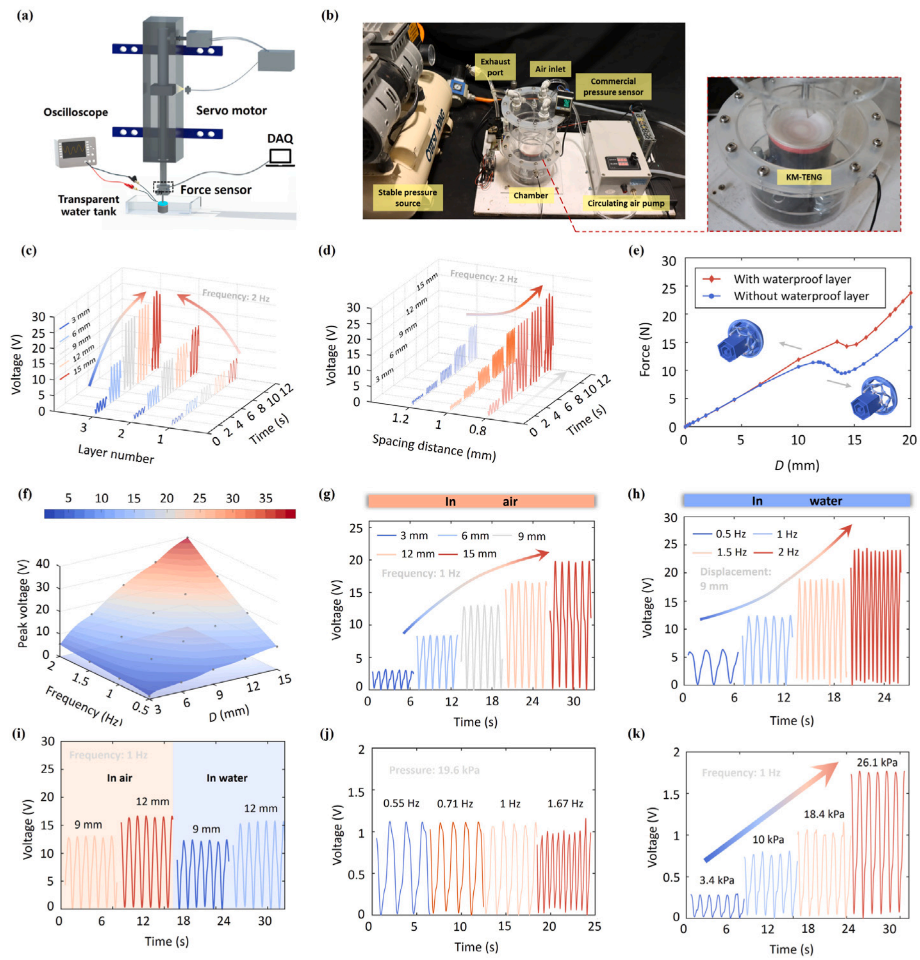

Fig. 3. Electrical performance of KM-TENG. (a) Servo motor force feedback system measurement setup. (b) Dynamic fluid pressure measurement setup. (c) Optimization of layer number of the load-electricity conversion component on output voltages. (d) Optimization of spacing distance of the load-electricity conversion component on output voltages. (e) Effect of waterproof layer on force–displacement relationship. (f) Peak V_{oc} of the encapsulated KM-TENG under vertical mechanical impact force cyclic loading at different deformations and frequencies in air. (g) Output voltages under vertical mechanical impact force cyclic loading at different deformations in air. (h) Output voltages under vertical mechanical impact force cyclic loading at different frequencies in water. (i) Comparison of V_{oc} for KM-TENG under vertical mechanical impact force cyclic loading in air and water. (j) V_{oc} of KM-TENG under dynamic fluid pressure at different frequencies. (k) V_{oc} of KM-TENG under dynamic fluid pressure at different pressures.

more efficient conversion of mechanical deformation into changes in the dielectric contact area. The linear relationship observed in both air and water (sensitivity difference < 7%) validates the design efficacy of the waterproof layer.

The waterproof layer ensures minimal loss in load transmission in aqueous environments, as evidenced by the consistent mean R^2 (0.98 in air and 0.97 in water). Mechanically, Due to the deformability and stretchability of the kirigami part in the sensitive element, it can convert applied forces into reversible deformations. For dynamic fluid pressure sensing (Fig. 4c), the peak voltage exhibits a strong linear correlation with the peak pressure ($R^2 = 0.97$), with a sensitivity of 0.07 V/kPa. This response is enabled by the concentric annular corrugations of the waterproof layer, which converts the distributed fluid pressure into localized stresses on the kirigami part. The enclosure walls of the load-electricity conversion component facilitate controlled sliding under pressure. The observed sensitivity is consistent with the theoretical prediction that peak output voltages with the rate of change in pressure, as validated by consistent linearity at the tested pressure range (3.4–26.1 kPa).

The robustness of the KM-TENG is evaluated using the servo motor force feedback system. To simulate real seawater conditions, which feature temperature fluctuations, salinity, and suspended sediment concentrations compared to pure water, a series of tests are conducted using customized water samples. The output performance of the KM-TENG is evaluated across temperatures from 11.3°C to 32.1°C (Fig. 4d). The KM-TENG exhibits minimal variation in output, with a relative standard deviation of 2.71% and a linear temperature coefficient of 0.18%/°C. Although a slight decrease in peak V_{oc} occurs at the lowest temperature, the overall performance remains highly stable within this range. The observed fluctuations have a negligible impact on sensing accuracy, and any potential drift can be further compensated should higher precision be required. Taking account of typical seawater salinity ranges from 15 ‰ to 35 ‰, incremental salt concentrations (15 g, 20 g, 25 g, 30 g, and 35 g per kilogram of water) are introduced into pure water. The performance of KM-TENG in these saline solutions is presented in Fig. 4e, where the peak V_{oc} shows minor fluctuations under varying salinities. The voltage output of KM-TENG remains largely stable after the 96-hour continuous immersion test in water with a salinity of 35 ‰, confirming its reliable long-term waterproof performance under high-salinity conditions (see Note S9). Furthermore, sand particles with diameters of 50–100 μm are added to pure water in specific quantities to achieve suspended sediment concentrations (SSC) of 1 kg/m^3, 3 kg/m^3, 5 kg/m^3, 10 kg/m^3, and 15 kg/m^3. Normal conditions correspond to SSC below 3 kg/m^3, while higher SSC levels occur under extreme ocean dynamics, such as the

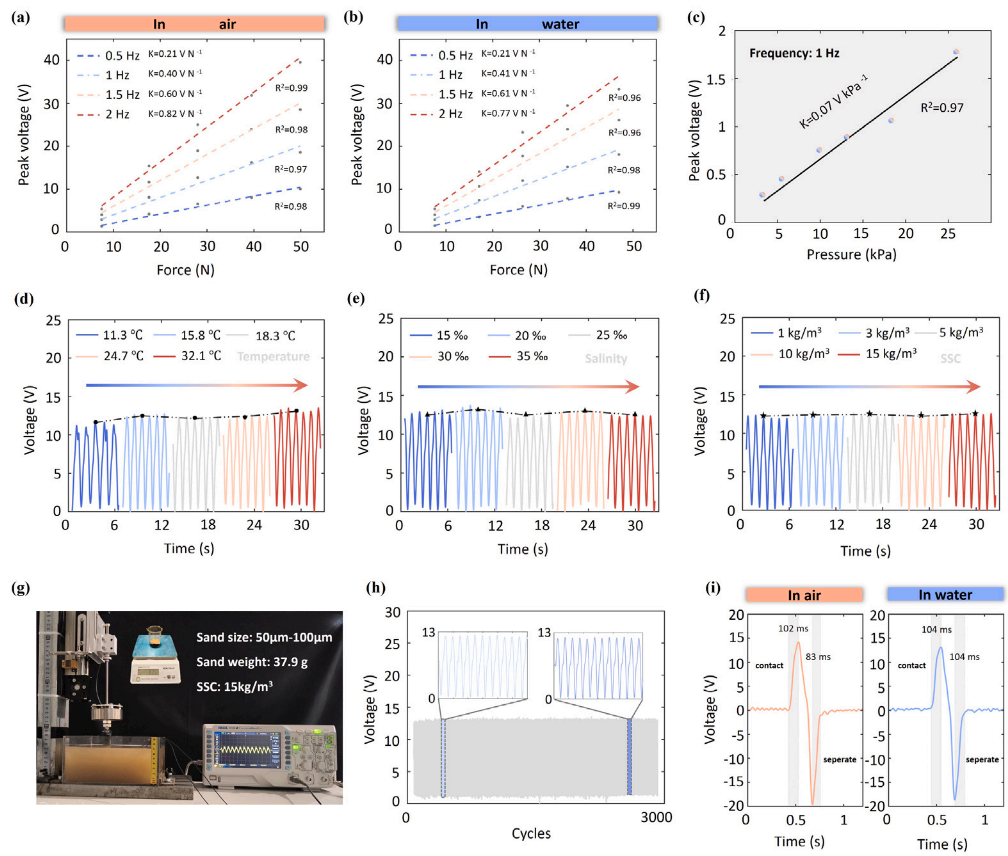

Fig. 4. Responses and robustness of KM-TENG. (a) Correlation of the vertical mechanical impact force and peak voltage in air. (b) Correlation of the vertical mechanical impact force and peak voltage in water. (c) Correlation of the dynamic fluid pressure and peak voltage. (d) V_{oc} of KM-TENG at different water temperatures. (e) V_{oc} of KM-TENG at different salinities of water. (f) V_{oc} of KM-TENG at different suspended sediment concentrations of water. (g) The image of KM-TENG working in sediment-laden water with SSC of 15 kg/m^3. (h) The repeatability test results of KM-TENG, with inserts displaying enlarged signals for different parts of the cycles. (i) Response speed of KM-TENG in air and water.

passage of typhoon. Fig. 4f illustrates the output voltages of KM-TENG in water with different levels of SSC, indicating that SSC within this range has a negligible effect on its performance. Fig. 4g shows the KM-TENG operating in sediment-laden water with an SSC of 15 kg/m^3, demonstrating its feasibility in high-SSC water. After 3000 cycles (50 min) at a deformation amplitude of 9 mm and a frequency of 1 Hz, V_{oc} remains stable at approximately 12.1 V (Fig. 4h). Furthermore, the response time of the KM-TENG in air and water is carefully measured at loading and unloading velocities of 50 mm/s and a deformation amplitude of 9 mm. During continuous loading and unloading, which involves one full cycle of the sensitive element sliding in and out of the load-electricity conversion component, the response time is depicted in Fig. 4i. In air, the response time is 102 ms during loading and 83 ms during unloading. In water, the response time increases slightly to 104 ms for both loading and unloading. These millisecond-scale response times highlight the rapid response ability of the KM-TENG and confirm its capacity to capture signals with millisecond temporal resolution.

3.4. Application of wave slamming pressure sensor

The KM-TENG has notable advantages, such as large deformation capacity, mechanical programmability, consistent performance in air and water, robustness under water with salinity and suspended sediment concentrations (SSC), durability, and millisecond-range temporal resolution, making it an ideal self-powered sensor for marine structure monitoring. Based on this sensor, a real-time wireless monitoring system for marine structures has been developed. The workflow of the system, as illustrated in Fig. 5a, encompasses signal generation, acquisition, transmission, and processing. The KM-TENG can capture cyclic loads, such as wave slamming pressures acting on marine structures, and convert mechanical inputs into electrical signals. These signals are then transmitted through wireless communication modules (e.g., LoRa) to terminal devices for real-time monitoring purposes. The wireless communication module based on the LoRa protocol (Fig. 5b) is composed of four components: (1) a signal conditioning module (CS1238) responsible for analog-to-digital conversion, (2) an STM32F103C8T6 microcontroller for data processing, (3) a LoRa transmitter and receiver (an ideal rate of 62500 bps and a range of 6 km), and (4) a 3400 mAh battery pack. The voltages acquired by the wireless communication module exhibit a linear relationship with the voltages measured by the oscilloscope (see Note S10). The terminal, a MATLAB-based host computer, receives the transmitted signals and processes them to enable visualization. Dedicated experiments in both indoor and outdoor environments demonstrate the high reliability of the LoRa-based wireless transmission system, showing consistent data integrity and low latency under varying conditions. The measured mean latency is 6.47 ms indoors and 6.33 ms outdoors (Table S11), with the marginally lower outdoor latency suggesting improved signal propagation efficiency due to reduced multipath reflection and electromagnetic interference. These results confirm the robustness and adaptability of the system across diverse operational environments.

In the 75 m × 1.8 m × 2 m wave flume experiment, periodic waves (0.12–0.16 m height) break over a flat bed of 0.2 m water depth after propagating down a 1:10 slope. The observed breaking wave height-to-depth ratio (H/h = 0.6) is lower than the theoretical threshold (H/h = 0.78). This discrepancy arises because seabed friction

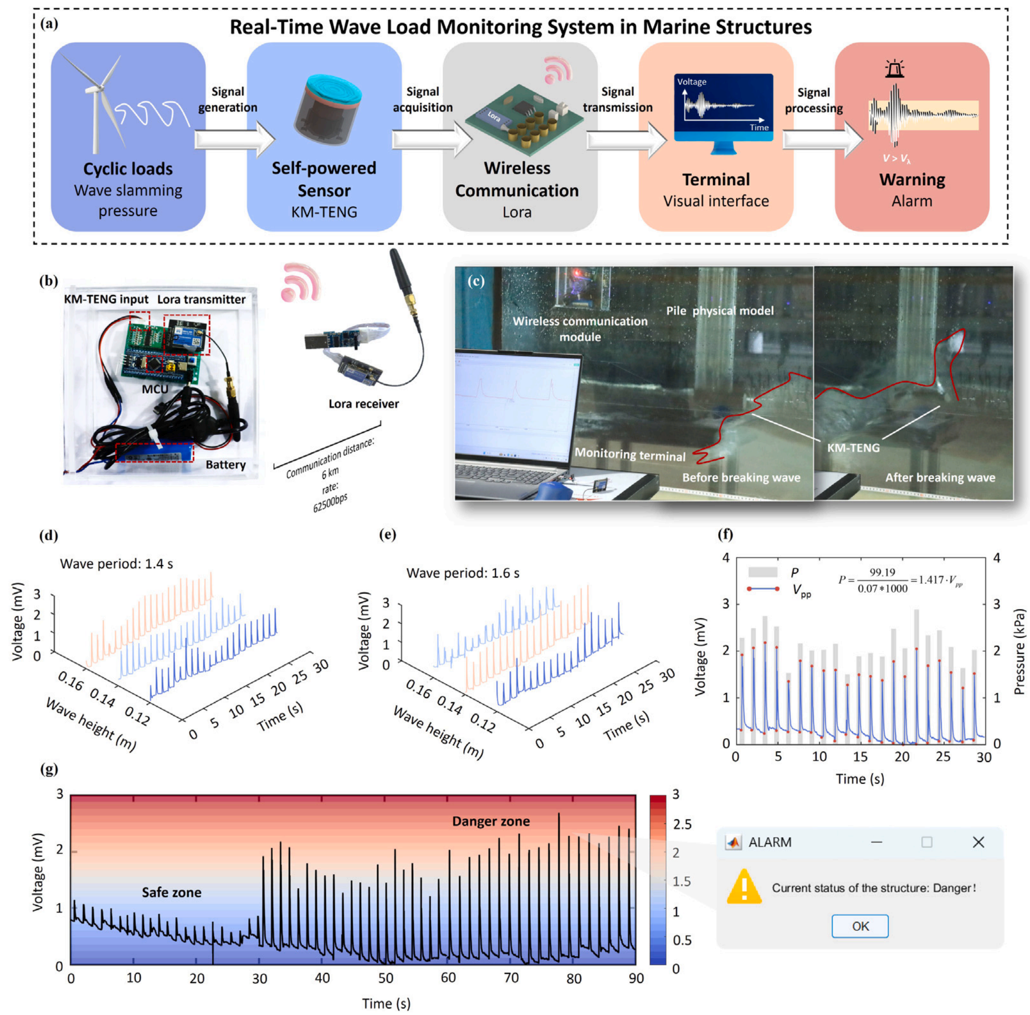

Fig. 5. Monitoring and early warning of wave slamming on marine structure (a) The workflow of the real-time wireless monitoring system. (b) Wireless communication module. (c) Monitoring of breaking wave slamming pressures in a large section wave tank. (d) Monitoring of breaking waves at different wave heights with wave period of 1.4 s. (e) Monitoring of breaking waves at different wave heights with wave period of 1.6 s. (f) The conversion of voltage amplitude V_{pp} to pressure value P. (g) Wave pressures on the pile in safe zone and danger zone.

and topographic shoaling synergistically enhance wave steepening in shallow water, making breaking conditions easier to achieve than in idealized theoretical models. The wave crest becomes vertical at the breaking point in the front of the pile model, generating splashing and transient pressures directed towards the pile model. The KM-TENG mounted 5 mm above the water surface senses the pressure changes, driving a physical-electrical conversion (see Fig. 5c). The concentric annular corrugations of waterproof layer transfers impact pressures to the kirigami part, inducing buckling deformation that enables the dielectric materials slide relatively between each other. This generates the voltage signals in Fig. 5d–e, with each output voltage directly aligned to wave breaking instants. Under the smallest wave condition tested (H=0.12 m, T=1.4 s), the monitoring system records a peak voltage of 0.88 mV versus a noise floor of 0.038 mV, resulting in an SNR of 27.3 dB (details in Note S11); larger waves yield even higher SNRs, demonstrating that the voltage signals possess a high signal-to-noise ratio suitable for reliable pressure monitoring in dynamic aquatic environments.

The output voltages generated through this process enable reliable pressure quantification based on a linear correlation derived from the air pressure generation device. This apparatus effectively reproduces key frequency characteristics of wave-induced pressures, as evidenced by the consistent output at 0.71 Hz in both air pressure tests (Fig. 3j) and wave flume experiments (Fig. 5d) under equivalent wave number conditions. Since the output voltage of the KM-TENG remains independent of wave frequency under constant pressure magnitude, the established linear relationship can be confidently used to invert pressure values from voltage signals. Thus, These signals enable precise conversion from voltage amplitude V_{pp} to pressure value P, e.g., a 1.61 mV bimodal peak translates to 2.282 kPa using the formula P = 1.417 V_{pp} (Fig. 5f). It is observed that wave heights exceeding 0.14 m result in significantly increased voltage outputs, which means a sharp increase in pressure. Visible structural shaking of the pile model is also observed alongside these changes. Therefore, the threshold is set at 2.834 kPa (equivalent to 2 mV) to demarcate the transition into hazardous conditions in Fig. 5g. For real-world applications, more comprehensive approaches including physical model tests and numerical simulations, which account for complex environmental factors such as suspended sediment concentration and wave–current interactions are needed. When the V_{pp} of KM-TENG exceeds this threshold (corresponding to increasing wave intensity and associated structural risk), the system triggers an alarm, with the user interface prompting for structural protection measures. The monitoring results show a reasonable agreement with reference values (see Note S12). This system offers effective strategies for dynamic wave load assessment and risk management of marine structures.

4. Conclusion

This study introduces a three-layer integrated compact cylindrical KM-TENG, which utilizes a kirigami metamaterial as the sensitive element to overcome the limitations of conventional marine sensors. The KM-TENG establishes a unique position in the sensing field due to the following key advantages: (1) Dual-parameter monitoring with high dynamic response. Unlike conventional sensors and most TENG-based sensors limited to single-parameter detection, KM-TENG simultaneously monitors both vertical mechanical impact forces and dynamic fluid impact pressures with a rapid response time of 102 ms. (2) Exceptional environmental robustness. KM-TENG operates reliably under a range of aquatic environmental parameters, including temperature variations, 11.3-32.1 °C, salinity up to 35‰, and suspended sediment concentration up to 15 kg/m^3, offering unparalleled suitability for real-world marine applications. (3) Tunable sensitivity and large deformation tolerance. The kirigami structure allows post-fabrication tuning of sensitivity and measurement range via geometric programming, and withstands large deformations up to 125%, providing both customizable sensing performance and extreme load resistance. (4) Self-powered and cost-effective operation. As a fully self-powered sensor, the KM-TENG eliminates the need for external power supplies, reducing deployment and maintenance costs. Its simple fabrication process enhances scalability for large-area marine monitoring networks. (5) Integrated with LoRa wireless communication and MATLAB-based host computer. The KM-TENG based monitoring system enables real-time structural health monitoring and early warning in marine environments.

Future work will pursue the key advances in KM-TENG marine monitoring: (1) Self-sustaining power provision for the wireless transmission module through ultra-low-power components, intermittent operation, and hybrid harvesting using wind/wave energy; (2) Expanded open-ocean testing under real-world constraints including extreme weather and complex tides to validate durability and refine deployment; (3) Continuous data collection and algorithm development using real wave patterns for adaptive thresholding and anomaly detection; (4) Scalable production via modular design and high-throughput manufacturing for cost-effective sensor arrays.

CRediT authorship contribution statement

Xiaolin Han: Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Wentao Li:** Validation, Investigation. **Xiyong Bai:** Investigation, Data curation. **Jiahao Liu:** Investigation, Data curation. **Zhiguo He:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Support information

Additional supporting information is available to complement this work, including detailed supplementary data, extended methods, and dynamic demonstrations, which are provided in separate files for clarity and accessibility.

1. Supplementary Document (Word format) : This file contains extended experimental procedures, additional data, and supplementary figures that further validate the findings presented in the main text.

2. Supporting Videos: Four videos are provided for illustration.

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.measurement.2025.119271.

Data availability

Data will be made available on request.

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